Liu Xingzhou

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Education

Sep 2023 – Jun 2027
Nanjing University of Information Science and Technology (NUIST)
School of Mathematics and Statistics
BSc in Mathematics and Applied Mathematics, expected June 2027

Sep 2027 (expected)
The Chinese University of Hong Kong (CUHK)
Division of Mathematics, Faculty of Science
MSc in Mathematics, expected start September 2027

Research Projects and Competitions

Apr 2025 – Apr 2026
Baltamatica for University Mathematics Teaching
Team lead, Jiangsu Provincial College Students' Innovation and Entrepreneurship Training Program

- Built a reusable lab curriculum and designed a five-stage process for evaluating platform features, identifying teaching needs, writing scripts, redesigning lessons, and assessing learning.
- Reached nearly 200 participants through questionnaires and interviews. Compared Baltamatica and MATLAB on syntax, function libraries, visualization, and compatibility.
- Wrote more than 50 scripts for 3D geometry, matrix transformations and eigenvalues, and finite-difference error visualization. Piloted the materials in three classes with about 30 students. Across more than 200 feedback responses, over 70% reported better comprehension with dynamic visualization.

May 2025
Marathon Planning and Economic Impact
Team lead, 10th Dimension Cup National College Mathematical Contest in Modeling
National Excellence Award

- Integrated weather, demographic, traffic, and registration data from 11 cities with spatial data for Xi'an. Processed and analyzed the data in Python and ArcGIS.
- Used TOPSIS to rank event months and K-means with the elbow method to group events into four tiers for frequency recommendations.
- Screened candidate endpoints by subway access and hotel capacity. Used DBSCAN, an improved ant colony algorithm, and breadth-first search to generate a 42 km route.
- Quantified shade in GIS, simulated traffic with cellular automata, and applied NSGA-II to balance shade coverage, traffic disruption, and route gradients.

Research Projects and Competitions (continued)

Feb 2025 – Apr 2025

HPV Vaccination Willingness in Nanjing
Team lead, National College Students' Statistical Modeling Competition

- Collected 525 valid responses from women aged 18–45 across 11 districts (Cronbach's $\alpha = 0.913$, KMO $= 0.916$). Integrated survey, literature, policy, and census data.
- Identified five factors associated with vaccination willingness using binary logistic regression. Applied the fuzzy analytic hierarchy process to weight 25 indicators across knowledge, socioeconomic, and policy dimensions.
- Proposed interventions for vaccine supply, subsidies, outreach, and market regulation.

Academic Exchange and Campus Activities

Jul 2024 – Aug 2024

Summer Academic Program, The University of Tokyo
Studied environmental governance, disaster prevention, aging societies, and education in Japan through lectures, seminars, and field visits. Visited physics laboratories to learn about nuclear fusion and experimental physics research.

Sep 2023 – Sep 2024

Arts and Culture Department, Student Union
School of Mathematics and Statistics, NUIST
Organized more than 10 campus events, including the Mathematics Culture Festival, Campus Singing Competition, and Campus Music Festival. Served as a peer mentor in the Huayang Classroom program.

Honors and Awards

Dec 2024, 2025 — **Second-Class Scholarship**, NUIST

Dec 2024, 2025 — **Outstanding Student**, NUIST

Nov 2025 — **Third Prize**, Chinese Mathematics Competitions for College Students (Jiangsu Division)

Jun 2025 — **Second Prize**, NUIST Mathematical Modeling Competition

Skills and Certifications

Computing — Python, MATLAB, LaTeX, Microsoft Office

English — IELTS 6.5, CET-6 439, CET-4 491

Certifications — National Computer Rank Examination (NCRE), Level II
Putonghua Proficiency Test, Level 2-A